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An exploratory feasibility study of adaptive assessment: a multivariate, data-driven approach to formative evaluation

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Introduction: Adaptive assessment has emerged as a promising approach for addressing learner heterogeneity in digital and online education, yet many existing systems rely on predefined psychometric assumptions that inadequately capture the complex, non-linear relationships among learner characteristics, engagement patterns, and academic performance. This study reports an exploratory feasibility study of a data-driven framework for personalised adaptive question recommendation, designed to estimate anticipated item difficulty using a multivariate set of learner indicators comprising prior academic achievement, engagement levels, study behaviours, and prior course experience. Because the anticipated-difficulty label is derived from four of these same learner indicators, the framework is best understood as a proof-of-concept for recovering a theoretically motivated scoring rule rather than as an independently validated measure of difficulty.
Methods: To explore this proof-of-concept, an exploratory research design was adopted using an authentic dataset of 71 undergraduate students enrolled across two courses at a higher education institution in Saudi Arabia. A multilayer perceptron (MLP) model was employed to capture non-linear interactions among learner variables, with preprocessing safeguards including ordinal-preserving feature encoding, LOWESS-based outlier suppression, and stratification-controlled oversampling applied exclusively to the training partition. Model performance was evaluated using Repeated Stratified K-Fold cross-validation across nine classifiers, with holdout test evaluation and post hoc feature influence analysis conducted to support pedagogical interpretability.
Results: The proposed MLP (RMSprop) achieved 93.33% test accuracy (Wilson 95% CI: 70.2%–98.8%, n = 15), demonstrating stable predictive performance and generalisation beyond training data despite sample size constraints. However, the small holdout sample, and the single observation underlying the Hard-class result in particular, mean these figures should be read as indicative rather than precise. Feature analysis indicated that learner engagement and prior academic performance were the most influential contributors to anticipated difficulty estimation, while misclassification patterns were primarily confined to adjacent difficulty levels, reflecting the inherent continuity of learner challenge.

Discussion: The findings suggest that data-driven adaptive mechanisms can complement traditional assessment approaches by providing context-sensitive, learner-relative difficulty estimation aligned with formative assessment principles. Fairness was not empirically evaluated in this study and is identified as a direction for future work rather than a current contribution. While exploratory in nature, this work contributes to bridging the gap between machine learning–based modelling and principled educational assessment design, offering a foundation for future large-scale and multi-context investigations.

KEYWORDS

adaptive assessment, AI in education, multi-layer perceptron (MLP), online learning, performance prediction, personalised learning, student engagement

Introduction

Educational assessment plays a central role in shaping learning experiences, instructional decisions, and learner motivation (Black and Wiliam, 1998; Sadler, 1989). Traditional assessment approaches, particularly those based on static item difficulty and uniform testing conditions, often struggle to accommodate the diversity of learners' backgrounds, abilities, and engagement patterns. As a result, learners may experience assessments that are either overly challenging or insufficiently stimulating, both of which can negatively affect learning outcomes and persistence.

Adaptive assessment has been proposed as a means to address these challenges by dynamically adjusting assessment characteristics in response to learner performance and behaviour. Within educational evaluation research, adaptive assessment is closely linked to formative assessment principles, which emphasise timely feedback, appropriate challenge, and support for learning progression rather than summative judgement alone. Established psychometric approaches, such as Item Response Theory (IRT) and Rasch models, have provided robust foundations for adaptive testing by modelling item difficulty and learner ability under well-defined assumptions (Embretson and Reise, 2000; Lord, 1980). However, these models typically rely on linear or monotonic relationships and may be limited in their ability to incorporate rich, behavioural indicators such as engagement intensity, study patterns, or learning context.

Recent advances in learning analytics and educational data mining have introduced machine learning techniques as alternative tools for modelling complex educational phenomena. Neural networks, in particular, offer the capacity to capture non-linear interactions among multiple learner variables. Despite this potential, their application in adaptive assessment raises important questions regarding interpretability, validity, and alignment with educational evaluation principles. High predictive accuracy alone does not guarantee meaningful or fair assessment decisions, especially when models are trained on limited or context-specific data (Wang and Asif, 2025). To provide a more comprehensive evaluation, the present study compares neural network performance against a range of traditional classifiers, including logistic regression, decision trees, and support vector machines, recognising that simpler linear models may, in certain feature spaces, achieve competitive performance and offer complementary interpretability advantages.

The present study addresses this gap by reporting an exploratory feasibility study of a data-driven adaptive assessment framework that integrates learner academic and engagement indicators to estimate anticipated difficulty levels. Rather than claiming superiority over traditional psychometric methods, the framework is positioned as a complementary approach that may support formative assessment objectives in contexts where rich behavioural data are available. A central methodological caveat, central enough that it is introduced here rather than deferred to the Limitations, is that the target variable (Anticipated Difficulty) is not an independent, empirically measured construct: it is computed directly from four of the five learner indicators that are subsequently used as model inputs. This produces a circular, near-tautological relationship between predictors and outcome, in which a classifier can achieve high accuracy largely by recovering the known scoring rule rather than by learning a genuinely independent predictive relationship. While the scoring rule is grounded in formative assessment theory, it is a theoretically derived proxy rather than a validated measure of difficulty, and the high accuracy figures reported in this study should be interpreted in that light. Disciplinarily, this study is situated at the intersection of educational evaluation research and learning analytics, drawing upon the formative assessment literature (Black and Wiliam, 1998; Sadler, 1989) and the adaptive testing tradition grounded in psychometric theory (Embretson and Reise, 2000; Lord, 1980).

The contributions of this study are threefold. First, it introduces an adaptive assessment framework that incorporates multiple learner-related indicators beyond traditional performance measures. Second, it demonstrates how non-linear modelling techniques can be evaluated using cross-validation while maintaining pedagogical interpretability. Third, it situates machine learning–based adaptation within the broader discourse of educational evaluation by discussing implications for construct validity, the deferred evaluation of fairness, and formative use. By adopting this perspective, the study seeks to bridge the gap between data-driven modelling and principled educational assessment design.

This study addresses the following research questions:

RQ1: Can learner engagement and academic indicators jointly predict anticipated assessment difficulty?

RQ2: How stable is the predictive performance of the proposed model under cross-validation?

RQ3: To what extent do different learner indicators contribute to adaptive difficulty estimation?

Literature review

This study draws upon three primary disciplinary fields: educational assessment theory, learning analytics, and machine learning. As an inherently cross-disciplinary contribution, it engages with debates in each of these areas while seeking to connect them through a coherent evaluative framework. The review that follows is organised accordingly, beginning with the foundational principles of educational assessment before addressing advances in learning analytics and machine learning-based adaptive systems.

Educational assessment plays a foundational role in shaping learning trajectories, instructional decisions, and learner motivation. Within formal educational settings, assessment serves not only as a summative measure of achievement but, increasingly, as a formative mechanism for guiding learning progression. Seminal contributions by Black and Wiliam (1998) established that formative assessment, when characterised by timely feedback, appropriate challenge, and responsiveness to learner readiness, produces measurable gains in student achievement. Sadler (1989) further articulated the conditions under which feedback supports learning, emphasising that learners must be able to close the gap between their current and desired performance. These foundational principles underpin the present study's conceptualisation of anticipated difficulty as a learner-relative construct rather than a fixed property of an assessment item.

Traditional adaptive testing has been formalised through psychometric frameworks, most notably Item Response Theory (IRT) and the Rasch model, which model the probability of a correct response as a function of both item difficulty and learner ability (Embretson and Reise, 2000; Lord, 1980). These approaches have provided rigorous foundations for large-scale adaptive assessments, enabling principled item selection based on probabilistic ability estimates. However, their application typically requires substantial item-response calibration data, assumes unidimensional latent traits, and relies on monotonic item-response functions that may inadequately capture the complexity of learner behaviour in modern digital learning environments. As a result, there is growing interest in complementary, data-driven approaches that can operate on richer behavioural indicators without requiring large-scale psychometric calibration.

Recent developments in artificial intelligence and machine learning present promising avenues for addressing these limitations. Adaptive assessments are gaining traction for their ability to adjust question difficulty based on student behaviour and performance indicators. Halkiopoulos and Gkintoni (2024) promote AI-driven adaptive learning environments grounded in cognitive neuropsychology, demonstrating that engagement intensity and prior academic performance are among the most significant predictors of a learner's capacity to manage assessment challenge, a finding that directly informs the feature weighting adopted in the present study. Almalawi et al. (2024) demonstrate that machine learning techniques, including support vector machines, artificial neural networks, and decision trees, surpass conventional evaluation models in predicting student performance, with GPA and engagement identified as particularly strong predictors of assessment readiness. Alvarez-Icaza et al. (2024) further illustrate that integrating varied learner characteristics, including sociodemographic contexts, behavioural tendencies, and personal attributes, improves the accuracy of engagement and performance forecasting, highlighting the necessity of constructing detailed learner profiles to inform the design of adaptive educational systems.

The capacity of adaptive systems to support learning outcomes beyond prediction accuracy has also been empirically demonstrated. Christian et al. (2024) found that students who undertook adaptive assessments reported higher mean scores on retention measures and higher levels of engagement than counterparts assessed through traditional methods, even when individual item recommendations were not perfectly calibrated, a finding that contextualises the present study's misclassification patterns within adjacent difficulty levels as educationally tolerable rather than critically harmful. Obergassel et al. (2025) further established that adaptive assessments, which dynamically adjust difficulty based on learners' current behaviour and proficiency, enhance learning quality by maintaining optimal cognitive load and providing feedback aligned with learner readiness, with calibration errors tending to occur at category boundaries rather than across the full difficulty range.

Several studies have explored AI-driven solutions to improve assessment efficacy and personalisation across varied educational contexts. Abrar et al. (2025) explore the use of machine learning for adaptive, context-aware personalisation in virtual learning environments. Hamzaoui et al. (2024) present a hybrid AI system that evaluates instructional videos and autonomously formulates questions in accordance with learning objectives. Liu et al. (2024) examine generative AI methodologies for the dynamic curation of material based on learner interactions. Joseph (2025) compares standard evaluations with AI-based adaptive assessments, noting substantial enhancements in response accuracy and personalisation. Bhattacharya et al. (2024) propose a real-time ICT-based learning analytics system designed to monitor student engagement and classroom performance in blended learning environments, while Salloum et al. (2025) investigate emotion-aware AI systems that modify online assessments based on learners' emotional states. Bhowmick et al. (2023) present AI-augmented question creation tools designed for particular knowledge domains, and Ouyang et al. (2023) utilise AI analytics to assess student engagement and performance in engineering education. Collectively, these studies highlight the breadth of AI-enhanced assessment research; however, they exhibit significant disparities in data sources, scalability, and explainability, and frequently concentrate on academic attributes while overlooking the integration of behavioural indicators, a gap the present study directly addresses through its multivariate learner-profile approach.

Multilayer perceptron neural networks have attracted particular attention within adaptive educational modelling due to their capacity to capture non-linear interactions among learner variables. Bacanin et al. (2022) employ an Enhanced Brainstorm Optimisation method to train MLPs, demonstrating accelerated convergence and improved prediction accuracy, while also utilising the Bat Algorithm for MLP weight optimisation to strengthen generalisation and reduce susceptibility to local minima. Stankovic et al. (2022) present a hybridised Arithmetic Optimisation

Algorithm designed to optimise MLP networks, demonstrating improved adaptability when processing dynamic learner data. Cuk et al. (2021) devised a hybrid training methodology integrating genetic algorithms and artificial bee colony optimisation, enabling MLPs to manage intricate non-linear feature interactions within complex educational datasets. Wang and Asif (2025) address high-dimensional learning environments through metaheuristic-optimised MLPs for educational assessment, calling for integrative directions that combine data-driven flexibility with psychometric rigour, a direction the present study advances through its comparison of MLP performance against traditional classifiers and its discussion of hybrid IRT integration. While these optimisation-focused methodologies advance algorithmic efficiency, they predominantly emphasise convergence performance and neglect engagement integration or pedagogical interpretability, crucial elements for effective implementation in real educational settings.

The present study responds to this gap by employing a compact, regularised MLP architecture with standard optimisers and interpretable feature encoding, prioritising transparency and educational alignment over algorithmic sophistication.

Taken together, the reviewed literature establishes a clear trajectory: from psychometric foundations through machine learning-based prediction toward integrated, learner-centred adaptive frameworks that balance predictive performance with validity, fairness, and pedagogical relevance. The present study is positioned at this intersection, contributing an exploratory, feasibility-oriented, data-driven adaptive assessment framework that is explicitly grounded in formative assessment principles and evaluated with transparency regarding its methodological constraints, including the construct-feature overlap between its target variable and model inputs.

Methodology

Research design and perspective

This study adopts an exploratory, data-driven design to construct a personalised adaptive question recommendation system for undergraduate students. The system classifies each student's anticipated assessment difficulty as Easy, Moderate, or Hard using a set of academic and behavioural learner indicators and subsequently retrieves a matched question from a validated question bank aligned with the student's course, preferred assessment type, and predicted difficulty level.

The study is exploratory in nature and is situated within the context of formative assessment and learning analytics. It does not seek to establish causal relationships but to demonstrate the feasibility of a multivariate, machine-learning-based approach to adaptive difficulty estimation. The research addresses three questions: (1) whether learner engagement and academic indicators can jointly predict anticipated assessment difficulty; (2) how stable model performance is under cross-validation; and (3) to what extent different learner indicators contribute to difficulty estimation (Figure 1).

Engagement_Level undergoes the same mapping; (iv) Study_Hours is coerced to numeric, with unparseable entries set to NaN; (v) Taken_Course_Before is coerced to an integer with missing values defaulting to 0; (vi) any row with a remaining missing value across these four columns is dropped. After cleaning, 71 complete student records are retained (Figure 1).

Participants and context

The dataset was collected from 71 undergraduate students enrolled in two courses, 231-CCS-4 and 371-CCS-3, at a higher education institution during a regular academic term. Participation data comprised academic records (cumulative GPA and prior course outcomes) and self-reported learning behaviour indicators collected through an institutional survey instrument. All data were anonymised before analysis.

Data sources

Student Survey (Supplementary Appendix Table 2). The survey file contains one row per student with seven columns: Student_Id, Gpa_Level, Course_Name, Preferred_Assessment, Engagement_Level, Study_Hours, and Taken_Course_Before. On loading, the following cleaning operations are applied: (i) all string columns are whitespace-stripped; (ii) Gpa_Level is mapped from numeric codes {1, 2, 3} to the string labels {Low, Moderate, High}; (iii) Question Bank (Supplementary Appendix Table 1). The question bank contains 24 rows covering every combination of 2 courses × 4 assessment types × 3 difficulty levels. After loading, all string columns are whitespace-stripped and the label "Medium" is normalised to "Moderate". A pipeline assertion confirms that the resulting unique difficulty levels are exactly {Easy, Moderate, Hard} before any further processing proceeds. The question bank is used exclusively at the recommendation stage and plays no role in model training or evaluation (Figure 1).

Target variable

Anticipated difficulty

The target variable represents the anticipated difficulty of an assessment item for a given student. It is not treated as an intrinsic property of the item but as a learner-relative estimation, a construct reflecting how challenging an item is expected to be given the student's academic background and engagement patterns. This conceptualisation aligns with formative assessment principles, where the goal is to maintain learners within an appropriate challenge zone rather than to assign fixed difficulty labels independent of context.

The target label is derived using a scoring rule applied to each student's profile. A composite score is computed as:

$$\text{score} = \text{GPA_Level_num} + \text{Engagement_Level_num} + \text{study bonus} + \text{Taken_Course_Before}$$

where GPA_Level_num and Engagement_Level_num map the string categories Low, Moderate, and High to the integers 1, 2, and 3, respectively; study_bonus equals 1 if the student's Study_Hours exceeds the dataset median (2.0 h) and 0

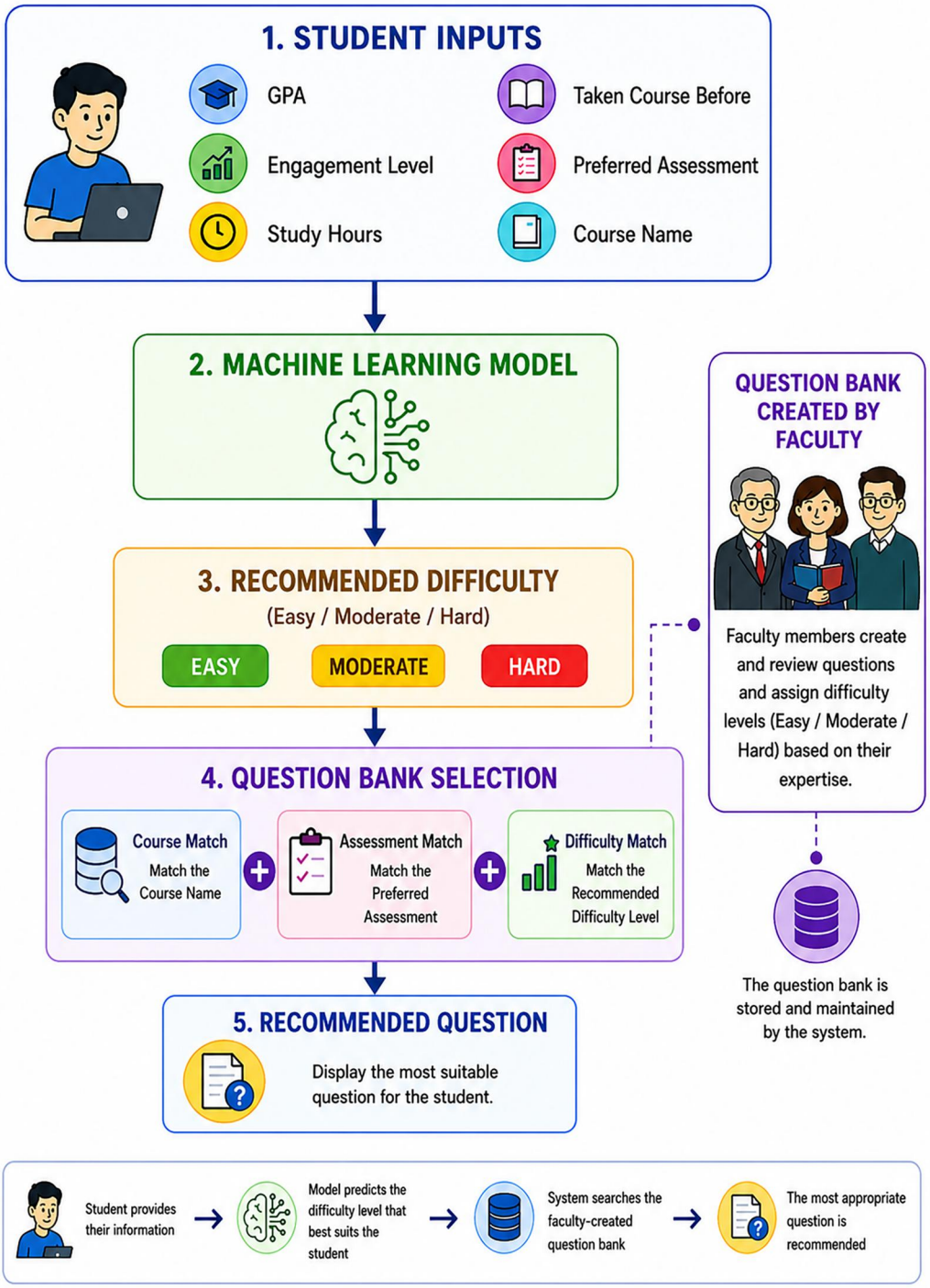


FIGURE 1
The study framework.

otherwise; and Taken_Course_Before is 0 or 1. The maximum achievable score is therefore 8. Score thresholds used to assign the Easy, Moderate, and Hard difficulty labels are listed in Table 1.

Applied to the 71-student dataset, this rule produces 18 Easy, 47 Moderate, and 6 Hard students, a distribution that is substantially more balanced than the earlier BASE_MAPPING rule, which assigned Hard to only 1 student and led to 49 near-identical oversampled copies. The scoring rule is the primary label-derivation method; the pipeline also supports real post-attempt student difficulty ratings via the REAL_LABELS_PATH parameter, which would replace the rule entirely when genuine self-rating data becomes available.

Stratified train

Test split

The 71-student dataset is partitioned into training and test sets before any oversampling is applied (Table 2). This ordering is a

critical methodological safeguard: splitting first ensures that the test set contains only real, never-seen student records, preventing the artificial train-test overlap that arises when oversampling is applied to the full dataset before splitting. In the previous version of this pipeline, oversampling preceded splitting, producing an estimated 72% feature-level overlap between training and test rows.

Stratification preserves the Easy/Moderate/Hard class proportions from the full dataset in both partitions. The test set of 15 rows is held out completely and is not used at any point during preprocessing, feature encoding, cross-validation, or hyperparameter tuning.

Training-set oversampling

The training partition (56 rows) is class-imbalanced: the Moderate class holds the majority of real students (approximately 37) while Hard holds approximately 5. To expose each classifier to balanced class representations during training, random oversampling with replacement (sklearn.utils.resample) is applied exclusively to the training set. Each class is up-sampled to match the count of the largest training class (37 rows), yielding 111 balanced training rows, 37 per class. Per-class oversampling counts are listed in Table 3.

The test set of 15 real student rows is never touched by the oversampler. Because the Hard class has only five real training students, a portion of its 37 training rows are duplicates; the pipeline reports the exact duplicate count per class at runtime for transparency.

TABLE 1 Difficulty labels and score thresholds.

Score threshold	Assigned label
Score ≥ 6	Hard
Score ≥ 4	Moderate
Score < 4	Easy

TABLE 2 Dataset partitioning details.

Parameter	Value
Method	sklearn train_test_split with stratify = y
Test fraction	20% → 15 real student rows
Train fraction	80% → 56 real student rows (before oversampling)
Random state	42 (fixed for full reproducibility)
Test class distribution	Easy: 4 | Moderate: 10 | Hard: 1

TABLE 3 Dataset partitioning details.

Class	Real training rows → Balanced rows
Easy	~14 → 37 rows
Moderate	~37 → 37 rows (majority class, no duplication)
Hard	~5 → 37 rows

Outlier suppression

Lowess smoothing

The Study_Hours column contains extreme outliers in the original survey data. A student self-reporting 73 study hours per week inflates the Gaussian variance estimates used by Naive Bayes, skews the StandardScaler's mean and variance, and distorts Euclidean distances used by KNN and SVM. To suppress this effect, LOWESS (Locally Weighted Scatterplot Smoothing) is applied to Study_Hours on the training set only.

LOWESS fits a locally weighted regression at each observation using its nearest neighbours within a specified bandwidth. The result is a smooth trend estimate that substantially reduces the influence of extreme values while preserving the overall distributional shape. The bandwidth parameter frac = 0.4 means each observation is smoothed using 40% of the nearest training rows. LOWESS smoothing parameters are listed in Table 4. It should be noted that the smoothing axis used here is row order, which is an arbitrary index rather than a temporally or theoretically meaningful variable; LOWESS is therefore applied as a pragmatic outlier-suppression heuristic rather than as a method that reduces genuine variance in a principled sense. A more theoretically grounded smoothing approach would require a time-stamped or otherwise meaningful ordering variable, which was not available for this dataset.

Fitting LOWESS only on training indices and applying a percentile cap to the test set ensures that no information from the test distribution leaks into the training preprocessing step.

Feature engineering and preprocessing

Five raw survey features form the input to all classification models. A ColumnTransformer pipeline applies three encoding strategies in parallel, fitted exclusively on the training set and applied without re-fitting to the test set.

Ordinal encoding rationale

GPA_Level and Engagement_Level carry a natural rank order (Low < Moderate < High). Encoding them as integers 0, 1, 2 via OrdinalEncoder allows distance-based models (KNN, SVM) and linear models (Logistic Regression, MLP) to exploit this rank relationship directly. Using OneHotEncoder for these features, as

TABLE 4 LOWESS smoothing parameters.

Parameter	Value
Method	statsmodels.nonparametric.smoothers_lowess.lowess
Bandwidth (frac)	0.4
Applied to	Training set only (row indices 0 … n_train)
Test set treatment	Study_Hours clipped to 95th-percentile of raw training distribution (6.0 h)
Effect on variance	Raw std = 8.45 h → LOWESS std = 0.58 h

in the previous version, would treat the three levels as unordered categories, discarding pedagogically meaningful information.

One-hot encoding rationale

Preferred_Assessment is a nominal variable (Fill in the Blanks, Multiple Choice, Short Answer, True or False) with no inherent ordering. OneHotEncoder produces four binary indicator columns, preventing any spurious ordinal interpretation.

Resulting input dimension

The ColumnTransformer produces an 8-dimensional encoded feature vector per student: 2 numeric (scaled) + 2 ordinals (integer-encoded) + 4 one-hot (assessment type). This is the input to all nine classifiers. Feature encoding and preprocessing configuration details are listed in Table 5.

Classification models

Nine classifiers are trained and evaluated: seven traditional machine learning baselines and two proposed MLP neural network variants. All classifiers receive the same 8-dimensional scaled feature matrix. Where the model supports a class_weight parameter, "balanced" is specified to further mitigate residual imbalance after oversampling.

It should be noted that the nine classifiers were not subjected to an equivalent, systematic hyperparameter search; each was trained using the fixed configuration listed in Table 6, selected from common practice and standard library defaults rather than from a tuned search over each model's hyperparameter space. Consequently, the comparison reported in this study reflects representative, commonly used configurations rather than each classifier's tuned best-case performance, and the relative ranking of models should be interpreted with this asymmetry in mind.

TABLE 5 Feature encoding and preprocessing configuration.

Feature	Raw Column	Encoding	Role
Study Hours	Study_Hours	StandardScaler (z-score normalisation)	Numeric – continuous
Prior Course Exp.	Taken_Course_Before	StandardScaler (binary 0/1 preserved)	Numeric – binary flag
GPA Level	Gpa_Level → GPA_Level	OrdinalEncoder Low=0, Moderate=1, High=2	Ordinal – rank preserved
Engagement Level	Engagement_Level	OrdinalEncoder Low=0, Moderate=1, High=2	Ordinal – rank preserved
Preferred Assessment	Preferred_Assessment	OneHotEncoder → 4 binary columns	Nominal – no natural rank

MLP architecture

Both MLP variants share a single hidden-layer architecture in line with the paper's modelling approach of using a limited number of neurons to mitigate overfitting on a small dataset. The architecture is (Table 7):

With 88 training rows available to the MLP (after 20% internal validation split), the parameter-to-sample ratio is approximately 2.2, which is deliberately conservative. The L2 penalty ($\lambda = 0.01$) on the hidden layer imposes additional weight regularisation beyond the Dropout layer, reducing the risk of overfitting the duplicated Hard-class. Training rows (Figure 2).

MLP training procedure

Early stopping

The full set of MLP training parameters is listed in Table 8. Training halts when val_loss fails to improve for 20 consecutive epochs, after which the weights from the best validation

TABLE 6 Machine learning models and configurations.

Model	Key configuration
Decision Tree	splitter='random', class_weight='balanced', random_state=42
Random Forest	n_estimators=200, class_weight='balanced', random_state=42
SVM (Linear)	kernel='linear', C = default, probability = True, class_weight='balanced'
Naive Bayes	GaussianNB, var_smoothing=1×10^{-4}
K-Nearest Neighbours	n_neighbors=5
Logistic Regression	max_iter=1,000, class_weight='balanced', random_state=42
Gradient Boosting	GradientBoostingClassifier, random_state=42
MLP – RMSprop	Dense (16, ReLU, L2 = 0.01) → Dropout (0.2) → Dense (3, Softmax); RMSprop lr = 0.001, rho=0.95, momentum=0.9
MLP – Adam	Dense (16, ReLU, L2 = 0.01) → Dropout (0.2) → Dense (3, Softmax); Adam lr = 0.001

TABLE 7 MLP network architecture.

Layer	Specification
Input	8 neurons — one per encoded feature ($\mathbb{R}^8$)
Hidden	Dense (16), ReLU activation, L2 regularisation $\lambda = 0.01$
Dropout	Rate = 0.2 — applied after hidden layer, before output
Output	Dense (3), Softmax activation — one neuron per class ($\mathbb{R}^3$)
Loss function	Sparse categorical cross-entropy
Parameters	$8 \times 16 + 16 + 16 \times 3 + 3 = 195$ trainable parameters

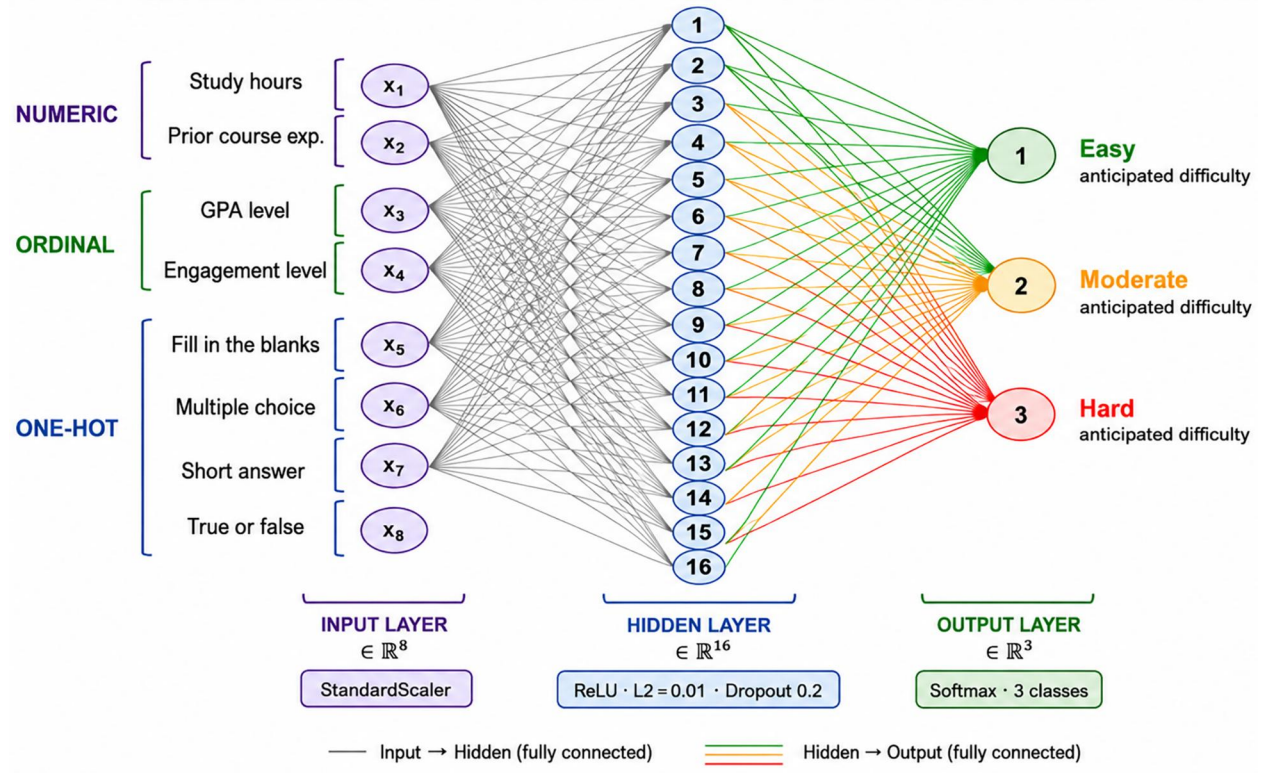


FIGURE 2
MLP structure.

TABLE 8 MLP training parameters.

Setting	Value
Maximum epochs	300
Batch size	16
Validation data	20% of training set (88 train/23 val), stratified, carved from X_train only
Early Stopping	monitor = val_loss, patience=20, restore_best_weights = True
ReduceLROnPlateau	monitor = val_loss, factor=0.5, patience=8, min_lr = 1×10^{-6}
Class weighting	compute_class_weight("balanced") passed to model.fit()
Random seed	42 — numpy and TensorFlow seeds reset before each model build

checkpoint are restored. The MLP (RMSprop) stopped at epoch 90; the MLP (Adam) stopped at epoch 251. In both cases, the validation split is drawn exclusively from X_train, so X_test is never seen during training or early stopping decisions.

ReduceLROnPlateau

If val_loss plateaus for 8 consecutive epochs, the current learning rate is halved (factor = 0.5) to a minimum of 1×10^{-6}. This allows progressively finer gradient steps near the minimum, which is particularly important for RMSprop convergence on the small, noisy oversampled training set.

Optimiser comparison

RMSprop (lr = 0.001, rho = 0.95, momentum = 0.9) normalises gradients per parameter using a running average of squared gradients and uses a momentum term to accumulate past update directions. This combination proved more effective on this feature space after LOWESS flattened the Study_Hours variance, achieving 93.33% test accuracy. Adam (lr = 0.001) converges faster per epoch but settles at a higher generalisation error (73.33% test accuracy), stopping at epoch 251.

Validity strategy

Repeated stratified K-fold cross-validation

To estimate the generalisation performance of the seven baseline classifiers, Repeated Stratified K-Fold Cross-Validation is applied on the training set exclusively. The test set is never used at this stage. The cross-validation configuration is detailed in Table 9, alongside the classifier configurations listed in Table 6.

Stratification in each fold preserves the training-set class proportions. Repeating the 5-fold scheme 10 times with different random partitions substantially reduces the variance of the mean accuracy estimate relative to a single k-fold run.

The ± standard deviation reported alongside each mean accuracy quantifies the stability of the estimate across folds.

Holdout test set evaluation

After cross-validation, each model is retrained on the full oversampled training set (111 rows) and evaluated on the held-out test set of 15 real student rows. The test set is touched exactly once per model. Both training accuracy and test accuracy are reported, together with the train-test gap (training accuracy−test accuracy), which is the primary diagnostic for overfitting or rule memorisation.

Performance metrics

The following metrics are computed for all models using sklearn.metrics functions with zero_division = 0:

- Accuracy: proportion of correctly classified test samples.
- Precision (weighted): for each class c, Precision(c) = TP(c)/(TP(c) + FP(c)). The weighted average weights each class precision by its support in the test set.
- Recall (weighted): Recall(c) = TP(c)/(TP(c) + FN(c)), weighted by support.
- F1-Score (weighted): harmonic mean of precision and recall, weighted by support. Serves as the primary metric given class imbalance in the 15-sample test set (4 Easy, 10 Moderate, 1 Hard).
- Train-Test Gap: training accuracy minus test accuracy. A small positive or negative gap with high test accuracy indicates genuine generalisation; a large positive gap indicates overfitting or rule memorisation.

Results

Dataset characteristics and preprocessing outcomes

The study dataset comprised 71 undergraduate students enrolled across two courses (231-CCS-4 and 371-CCS-3). Applying the paper's composite scoring rule to each student's profile yielded a class distribution of 18 Easy (25.4%), 47 Moderate (66.2%), and 6 Hard (8.5%), a moderately imbalanced three-class problem with Moderate as the dominant category.

TABLE 9 Cross-Validation configuration.

Parameter	Value
Method	sklearn RepeatedStratifiedKFold
Folds per repeat	5
Repeats	10
Total folds	50
Data used	X_train only — X_test never seen
Metrics reported	Mean accuracy, precision (weighted), recall (weighted), F1 (weighted), and accuracy standard deviation across 50 folds

The stratified 80/20 holdout split (applied before any oversampling) produced a training partition of 56 real students and a test partition of 15 real, never-seen students (Easy: 4, Moderate: 10, Hard: 1). Oversampling was then applied exclusively to the training set, upsampling each class to the majority-class count of 37 rows, yielding 111 balanced training rows (37 per class). The test set remained untouched throughout.

LOWESS smoothing reduced the standard deviation of Study_Hours in the training set from 8.45 h to 0.58 h, effectively eliminating the distorting influence of the 73-hour outlier without discarding any student record. Test-set Study_Hours values were capped at the 95th percentile of the raw training distribution (6.0 h). Feature encoding via the ColumnTransformer produced an 8-dimensional input vector per student: two z-score-normalised numeric features, two ordinally encoded categorical features preserving rank order, and four one-hot binary columns for Preferred_Assessment type.

Cross-Validation performance of baseline classifiers

Table 10 reports mean accuracy and standard deviation across 50 folds (Repeated Stratified 5 × 10 KFold, applied to the training set only). The results indicate that all seven baseline classifiers substantially exceed the random chance baseline of 33.33%, demonstrating that the five learner-profile features contain a genuine predictive signal for anticipated difficulty classification.

SVM (Linear) achieved the highest cross-validation accuracy (84.81% ± 5.89%), followed closely by Logistic Regression (82.92% ± 7.26%). Both are linear models with class-balanced weighting, suggesting that the relationship between the eight encoded learner features and anticipated difficulty is broadly linear in the transformed feature space. Gradient Boosting (79.84% ± 7.85%) and Random Forest (79.73% ± 7.04%) placed third and fourth, indicating that ensemble methods also capture useful signal but did not substantially outperform the linear models.

Decision Tree (78.84% ± 7.84%) and KNN (74.08% ± 8.20%) were competitive but more variable, with KNN exhibiting the largest standard deviation among all classifiers (8.20%), reflecting sensitivity to the specific fold composition, expected

TABLE 10 Cross-validation performance of baseline classifiers (repeated stratified 5 × 10 KFold, X_train only).

Model	CV Acc (%)	CV Prec (%)	CV Rec (%)	CV F1 (%)	Std (%)
SVM (Linear)	84.81	87.05	84.81	83.70	5.89
Logistic Regression	82.92	85.33	82.92	81.54	7.26
Gradient Boosting	79.84	81.26	79.84	78.97	7.85
Random Forest	79.73	81.05	79.73	79.13	7.04
Decision Tree	78.84	80.09	78.84	78.13	7.84
KNN	74.08	76.62	74.08	71.42	8.20
Naive Bayes	70.00	67.77	70.00	62.46	4.42

given the small training set size and distance-metric sensitivity to the oversampled Hard-class instances.

Naive Bayes recorded the lowest cross-validation accuracy (70.00% ± 4.42%). Its relatively small standard deviation reflects consistent, predictable, if modest, predictive capacity. The discrepancy between its CV precision (67.77%) and CV recall (70.00%) indicates class-level asymmetries in its probability estimates, likely caused by the Gaussian variance assumptions being disrupted by the skewed Study_Hours distribution even after LOWESS smoothing (Figure 3).

Holdout test set performance

Table 11 presents the holdout evaluation results for all nine models on the 15 real, unseen test students. The train–test gap column, defined as the difference between training accuracy and test accuracy, serves as the primary indicator of overfitting (Figure 4). Because the test set contains only 15 students, each accuracy figure in Table 11 carries a wide Wilson 95% confidence interval; for example, the 93.33% test accuracy achieved by MLP (RMSprop) and Logistic Regression corresponds to a Wilson 95% CI of approximately 70.2%–98.8%, and the 80.00% accuracy achieved by KNN corresponds to a Wilson 95% CI of approximately 54.8%–93.0%. These intervals indicate that, while the point estimates favour MLP (RMSprop) and Logistic Regression, the holdout sample is too small to establish statistically significant differences in accuracy between several of the top-performing models, and the reported accuracy figures should be read as indicative rather than precise.

MLP (RMSPROP): best overall model

The Proposed MLP Neural Network with RMSprop optimiser achieved the highest test accuracy of 93.33%, with a weighted precision of 94.67%, weighted recall of 93.33%, and weighted F1-score of 93.53%. Notably, the model's training accuracy was

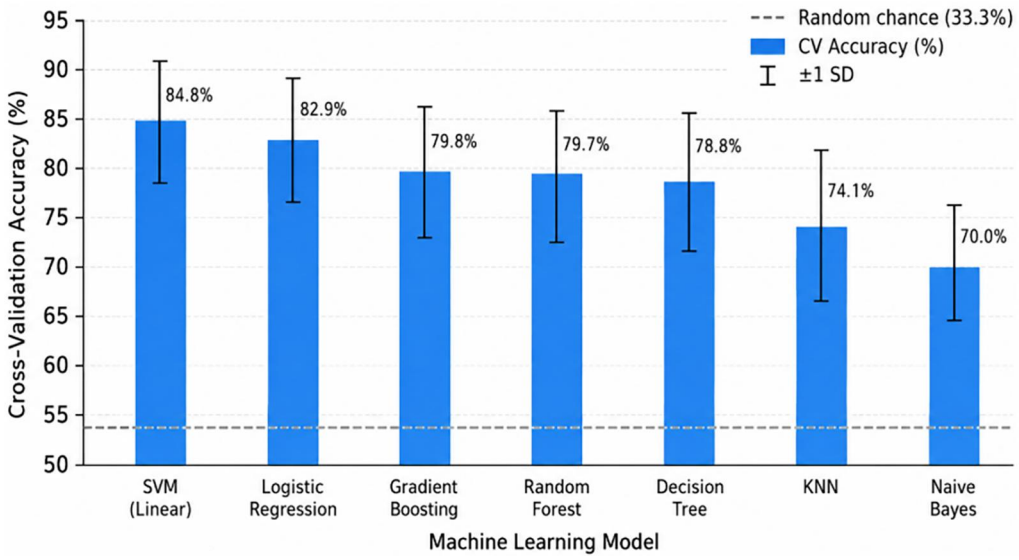


FIGURE 3
Cross-validation accuracy (Repeated Stratified 5 × 10 KFold, X_train only). Error bars show ±1 SD. Dashed line indicates random-chance baseline (33.3%).

TABLE 11 Holdout test set performance (15 real unseen students), sorted by test accuracy.

Model	Train (%)	Test (%)	Prec (%)	Rec (%)	F1 (%)	Gap (%)
MLP – RMSprop (proposed)	89.77	93.33	94.67	93.33	93.53	−3.56
Logistic Regression	90.09	93.33	94.67	93.33	93.53	−3.24
KNN	84.68	80.00	88.57	80.00	80.96	4.68
MLP – Adam (proposed)	85.23	73.33	86.67	73.33	74.44	11.89
Random Forest	100.00	73.33	73.57	73.33	71.25	26.67
SVM (Linear)	88.29	73.33	86.67	73.33	74.67	14.95
Gradient Boosting	100.00	66.67	71.85	66.67	69.12	33.33
Decision Tree	100.00	60.00	70.79	60.00	61.06	40.00
Naive Bayes	72.07	33.33	43.03	33.33	25.33	38.74

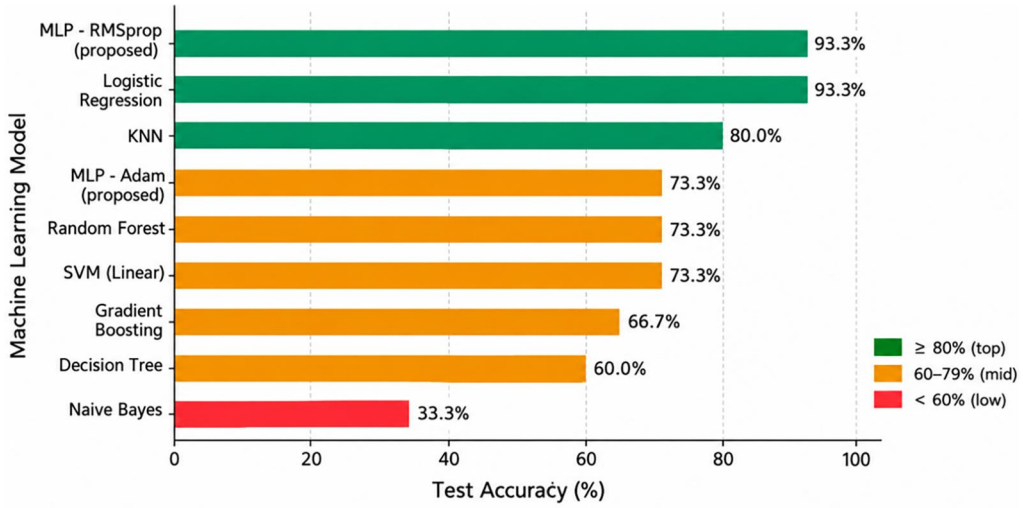


FIGURE 4
Holdout test set accuracy for all nine models (15 real, unseen students). Colours indicate performance tier: green ≥80%, amber 60%–79%, red <60%.

89.77%, producing a negative gap of −3.56%. A negative gap arises when EarlyStopping restores the best-checkpoint weights, which generalise marginally better to the held-out test set than the final training epoch would. The model converged at epoch 90 out of a maximum of 300, indicating efficient convergence under the RMSprop optimiser with momentum (rho = 0.95, momentum = 0.9) and the ReduceLROnPlateau scheduler.

At the class level, MLP (RMSprop) achieved perfect precision and recall on the Hard class (both 1.00), correctly classifying the single Hard test student. As this Hard-class result is based on a single test observation and a training subset in which most Hard-class rows are oversampled duplicates of a small number of real students, it does not permit inference about Hard-class performance at the population level and is reported here only as a descriptive observation. For the Moderate class, the most populated with 10 test instances, it achieved a precision of 1.00 and a recall of 0.90, misclassifying one Moderate student as Easy. For Easy, recall reached 1.00 (all four Easy students correctly identified) with a precision of 0.80, reflecting a single false-positive from the Moderate class. This misclassification pattern, confined to adjacent difficulty levels (Moderate ↔ Easy), is consistent with the inherent continuity of learner challenge and aligns with formative assessment theory, where boundary ambiguity between adjacent difficulty categories is expected.

Logistic regression: equal best on test set

Logistic Regression matched MLP (RMSprop) with 93.33% test accuracy, identical weighted precision (94.67%), recall (93.33%), and F1 (93.53%), and a near-zero gap of −3.24%. Its per-class performance was identical to MLP (RMSprop): all four Easy students were correctly identified, nine of ten Moderate students were correctly classified, and the single Hard student was perfectly classified. This result is noteworthy: a regularised linear model achieves state-of-the-art performance on this dataset, suggesting that the decision boundaries separating the three difficulty classes are largely linear in the 8-dimensional encoded feature space.

KNN: strong generalisation with minimal gap

K-Nearest Neighbours (k = 5) achieved 80.00% test accuracy with a small train–test gap of only 4.68%, indicating sound generalisation. KNN correctly classified all four Easy students and perfectly classified the single Hard student (precision = 1.00, recall = 1.00). Three Moderate students were misclassified as Easy, reflected in a Moderate recall of 0.70. The small gap is partly attributable to LOWESS smoothing, which reduced Study_Hours variance and thereby stabilised the Euclidean distances that KNN relies on.

MLP (Adam): moderate performance, slower convergence

The MLP with Adam optimiser achieved 73.33% test accuracy and stopped at epoch 251, substantially later than MLP (RMSprop), indicating slower convergence. Its train–test gap of 11.89% suggests moderate overfitting. While Adam correctly classified all Easy and Hard test students, four of ten Moderate students were misclassified as Easy, producing a Moderate recall of 0.60. The inferior performance of Adam relative to RMSprop is consistent with the hypothesis that RMSprop's per-parameter gradient normalisation provides more stable optimisation when Study_Hours has been smoothed by LOWESS.

Tree-based and ensemble models: overfitting pattern

Random Forest (73.33%), SVM (73.33%), Gradient Boosting (66.67%), and Decision Tree (60.00%) all exhibit substantial train–test gaps: 26.67%, 14.95%, 33.33%, and 40.00%, respectively. Decision Tree and Gradient Boosting both achieved perfect training accuracy (100.00%), indicating that they fully memorised the oversampled training set. Critically, all three tree-based models failed to classify the single Hard test student (Hard precision = 0.00, recall = 0.00), revealing that their overfit decision boundaries did not generalise to the minority class (Figure 5).

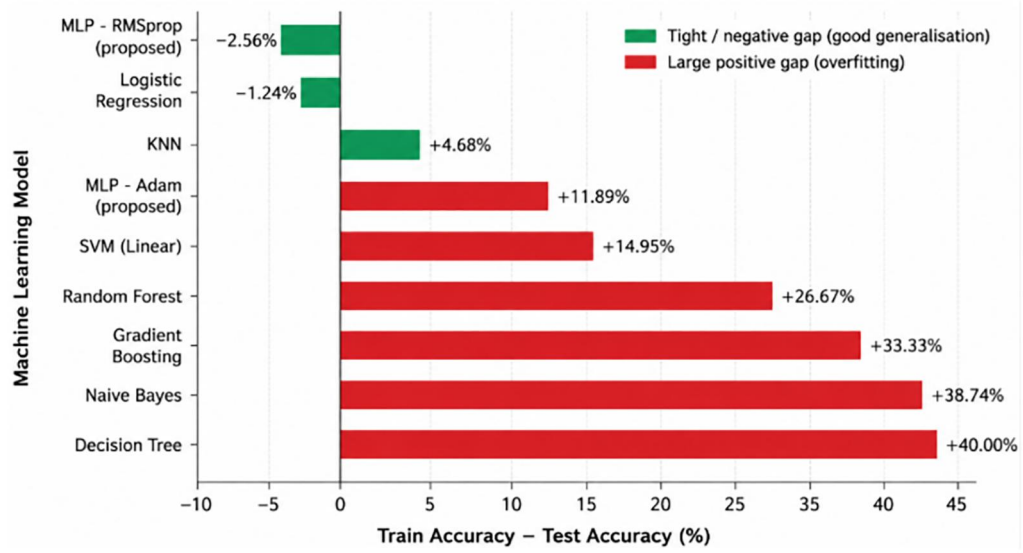


FIGURE 5
Train–test accuracy gap for all nine models. Negative or near-zero gaps (green) indicate good generalisation; large positive gaps (red) indicate overfitting.

Naive Bayes: lowest test performance

Naive Bayes achieved 33.33% test accuracy, the lowest among all nine models and approximately equal to random chance. At the class level, Naive Bayes correctly classified all four Easy students (recall = 1.00) but classified only one of ten Moderate students correctly (recall = 0.10) and failed on Hard (recall = 0.00). This pattern suggests the model defaulted predominantly to predicting Easy on the test set.

Per-class classification performance summary

Table 12 summarises per-class precision, recall, and F1-score across all nine models on the 15-student test set (Figure 6). As with Table 11, the Hard-class row for every model in Table 12 is derived from a single test student and should not be read as evidence of generalisable Hard-class performance at the population level.

Across models, performance on the Hard class is highly variable: MLP (RMSprop), Logistic Regression, KNN, and MLP (Adam) all achieved perfect Hard classification (F1 = 1.00), while Decision Tree, Random Forest, Gradient Boosting, and Naive Bayes achieved Hard F1 = 0.00. This dichotomy likely reflects the degree to which each model generalised beyond the oversampled training instances. Because both outcomes are drawn from a single Hard-class test student, this dichotomy should be read as a binary outcome on one case rather than as a reliable estimate of population-level Hard-class accuracy for any model.

The Moderate class, with 10 test instances, proved the most discriminating benchmark. MLP (RMSprop) and Logistic Regression achieved Moderate recall of 0.90 (one misclassification each), while most other models showed Moderate recall between 0.50 and 0.70, with Naive Bayes reaching only 0.10. The Easy class showed consistently high recall across all models ($\geq$ 0.75), suggesting that the Easy learner profile is reliably distinguishable from the other two categories in the encoded feature space.

Discussion

RQ1: can learner engagement and academic indicators jointly predict anticipated assessment difficulty?

The results provide affirmative evidence for RQ1. All nine classifiers substantially exceeded the random chance baseline of 33.33%, with the best models achieving 93.33% accuracy on a held-out sample of real, previously unseen students (Wilson 95% CI: 70.2%–98.8%). This indicates that the five learner-profile features, GPA level, engagement level, study hours, prior course experience, and preferred assessment type, contain meaningful predictive signals for the three-category difficulty classification task, although the width of this interval, driven by the small 15-student holdout sample, means the affirmative answer to RQ1 should be regarded as provisional rather than conclusive.

The strength of Logistic Regression (93.33% test accuracy) is particularly informative: a model that learns a simple linear boundary in the 8-dimensional feature space achieves state-of-the-art performance, suggesting that the relationship between the learner indicators and anticipated difficulty is substantially captured by linear combinations of those features. The positive contribution of GPA level and engagement level to higher difficulty predictions is consistent with established formative assessment literature, in which prior academic achievement and active engagement are among the strongest predictors of a learner's capacity to manage challenging material (Almalawi et al., 2024; Alvarez-Icaza et al., 2024).

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TABLE 12 Per-class classification performance on the 15-student holdout test set.

Model	Class	Precision	Recall	F1-Score
MLP – RMSprop	Easy	0.80	1.00	0.89
	Moderate	1.00	0.90	0.95
	Hard	1.00	1.00	1.00
Logistic Regression	Easy	0.80	1.00	0.89
	Moderate	1.00	0.90	0.95
	Hard	1.00	1.00	1.00
KNN	Easy	0.57	1.00	0.73
	Moderate	1.00	0.70	0.82
	Hard	1.00	1.00	1.00
MLP – Adam	Easy	0.50	1.00	0.67
	Moderate	1.00	0.60	0.75
	Hard	1.00	1.00	1.00
SVM (Linear)	Easy	0.67	1.00	0.80
	Moderate	1.00	0.60	0.75
	Hard	0.33	1.00	0.50
Random Forest	Easy	0.57	1.00	0.73
	Moderate	0.88	0.70	0.78
	Hard	0.00	0.00	0.00
Gradient Boosting	Easy	0.75	0.75	0.75
	Moderate	0.78	0.70	0.74
	Hard	0.00	0.00	0.00
Decision Tree	Easy	0.57	1.00	0.73
	Moderate	0.83	0.50	0.62
	Hard	0.00	0.00	0.00
Naive Bayes	Easy	0.36	1.00	0.53
	Moderate	0.50	0.10	0.17
	Hard	0.00	0.00	0.00

The study_bonus component of the scoring rule, awarding one point to students who study more than the dataset median, contributed to a more nuanced difficulty assignment that accounts for effort beyond raw academic standing. This aligns with Obergassel et al.'s (2025) finding that adaptive assessments, which account for current learner effort and proficiency, produce better-calibrated challenge levels than those relying solely on static ability measures.

RQ2: how stable is predictive performance under cross-validation?

The Repeated Stratified 5 × 10 K-fold cross-validation results (Table 10) demonstrate moderate-to-good stability across folds. SVM (Linear) and Logistic Regression achieved the lowest standard deviations among the higher-performing models (5.89% and 7.26%, respectively), indicating that their cross-validation accuracy estimates are robust to the specific composition of the training folds. This stability is consistent with the hypothesis that linear decision boundaries generalise reliably across the learner-profile feature space, irrespective of which subset of the 56 training students is held out in a given fold.

KNN exhibited the highest cross-validation standard deviation (8.20%), reflecting its sensitivity to the specific training instances included in each fold, particularly relevant given the imbalanced Hard class (approximately five real students). A single Hard student appearing or not appearing in a given training fold can substantially shift KNN's nearest-neighbour boundaries for that class. Naive Bayes, despite recording the lowest mean cross-validation accuracy (70.00%), exhibited the smallest standard deviation (4.42%), reflecting the parametric nature of Gaussian Naive Bayes, whose accuracy is consistently limited by the Gaussian feature-independence assumption.

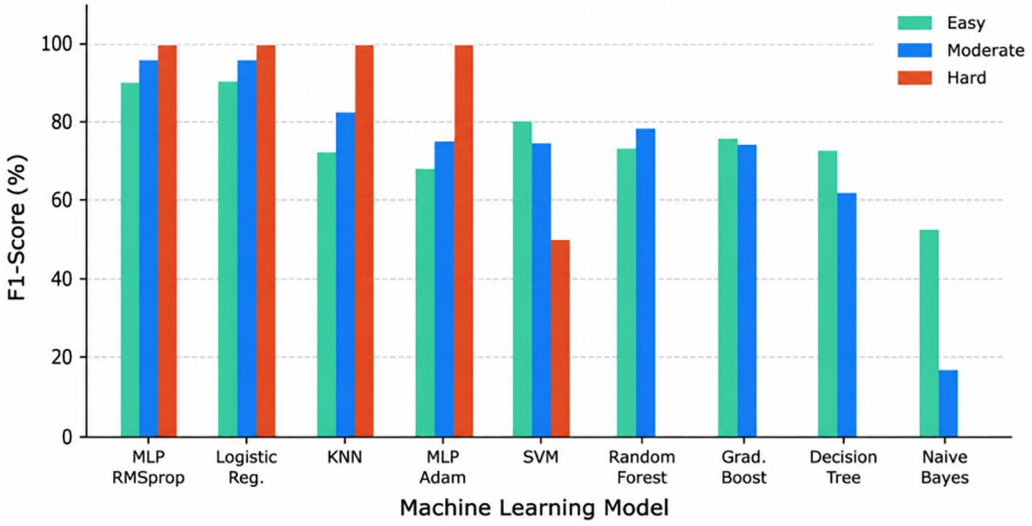


FIGURE 6
Per-class F1-scores on the holdout test set for all nine models. Easy = teal, Moderate = blue, Hard = coral.

RQ3: to what extent Do different learner indicators contribute to difficulty estimation?

The scoring rule assigns a maximum of three points to GPA level and engagement level combined (each contributing 1, 2, or 3 points), one point for studying above the dataset median, and one point for prior course experience. This weighting makes GPA and engagement the primary drivers of the difficulty classification, consistent with Halkiopoulos and Gkintoni's (2024) finding that engagement intensity and prior academic performance are the most significant predictors of a learner's capacity to manage assessment challenge.

The superior performance of models that exploit ordinal rank information, Logistic Regression and MLP (RMSprop), relative to models that treat features as unordered, further supports the conclusion that the ordinal structure of GPA and engagement contributes materially to predictive performance. The OrdinalEncoder choice (Low = 0, Moderate = 1, High = 2) allows these models to represent the monotonic relationship between academic standing and anticipated difficulty directly.

MLP performance relative to baseline classifiers

The proposed MLP (RMSprop) matched the best baseline classifier (Logistic Regression) at 93.33% test accuracy while achieving comparable precision, recall, and F1-score. The MLP uses only 195 trainable parameters, a deliberately conservative architecture, to model non-linear interactions among the eight encoded features. The ability of this compact network to match a well-regularised linear classifier suggests that, for this dataset, the non-linear interactions among learner indicators contribute only marginally to predictive performance beyond what is captured by linear terms.

The MLP (Adam) underperformed MLP (RMSprop) by 20 percentage points (73.33% vs. 93.33%), despite sharing the same architecture, training procedure, and regularisation. Adam's adaptive moment estimates accumulated large second-moment values early in training, causing the effective learning rate to decay rapidly and the model to converge to a suboptimal minimum. RMSprop with momentum (rho = 0.95, momentum = 0.9) maintained a more stable gradient normalisation and avoided premature convergence, producing a better-generalising model.

Misclassification patterns and educational implications

Across all models, misclassification errors were predominantly confined to adjacent difficulty categories: Easy students occasionally classified as Moderate, and Moderate students classified as Easy. No model, including the weakest performers, confused Easy with Hard or Hard with Easy in a systematic manner. This adjacency pattern is educationally meaningful: in formative assessment, a slight overestimation or underestimation of an adjacent difficulty level represents a minor calibration error rather than a fundamental mismatch.

This misclassification pattern is consistent with the paper's conceptualisation of anticipated difficulty as a continuous construct existing on a spectrum rather than as three sharply demarcated categories. It also aligns with Obergassel et al.'s (2025) observation that cognitive load-aware adaptive systems tend to produce errors at category boundaries rather than across the full difficulty range, and with Christian et al.'s (2024) finding that adaptive assessment systems achieve higher learner engagement even when individual item recommendations are not perfectly calibrated.

The Hard class presents a different pattern: models either classified the single Hard test student perfectly (F1 = 1.00, achieved by MLP-RMSprop, Logistic Regression, KNN, and MLP-Adam) or failed (F1 = 0.00, for Decision Tree, Random Forest, Gradient Boosting, and Naive Bayes). This bimodal outcome reflects the extremely limited Hard-class representation in both the training set (approximately five real students) and the test set (one student). This finding underscores the need for expanded Hard-class data collection in future iterations of the system.

Comparison with established adaptive assessment approaches

Traditional adaptive testing frameworks, including Item Response Theory (IRT) and Rasch models, establish item difficulty and learner ability through probabilistic models of item-response interactions, typically requiring large calibration samples (hundreds to thousands of responses per item). The present framework differs fundamentally in both its data requirements and its modelling assumptions: it estimates anticipated difficulty from a multivariate learner profile rather than from item-response history, and it employs a discriminative classifier rather than a generative psychometric model.

This distinction makes the proposed approach complementary to rather than competitive with IRT-based systems. In contexts where item-response data are not yet available, as in a newly deployed course or assessment platform, the learner-profile approach can provide an initial difficulty estimate that guides item selection before sufficient IRT calibration data have accumulated. This hybrid potential is consistent with the integrative direction called for by recent reviews of AI-enhanced adaptive assessment (Halkiopoulos and Gkintoni, 2024; Wang and Asif, 2025).

Validity, fairness, and pedagogical alignment

The use of a scoring rule to construct the target variable introduces a form of construct validity that is grounded in formative assessment theory: GPA, engagement, study effort, and prior experience are widely recognised as predictors of assessment readiness (Almalawi et al., 2024). However, because the rule is deterministic and uses the same features as the model input, the pipeline constitutes an approximation of genuine learner-difficulty alignment rather than a direct measure of it. The framework is therefore best understood as an adaptive support mechanism rather than a validated psychometric instrument.

From a design perspective, the multivariate modelling approach avoids over-reliance on any single indicator: no student is assigned a difficulty level based on GPA alone, engagement alone, or study hours alone. The composite score integrates multiple dimensions of learner context, which is intended to reduce, though it does not by itself rule out, the risk of systematic bias against students with lower academic records who compensate through high engagement or extensive study. It is important to note that no empirical fairness analysis, such as a comparison of model performance across demographic or socioeconomic subgroups, was conducted in this study; the multivariate design is a structural choice that may support fairer outcomes, not an evaluated fairness property of the resulting model. Future extensions of this framework should include explicit fairness audits across demographic subgroups, particularly given that GPA and engagement patterns may correlate with socioeconomic and linguistic background factors not captured in the current dataset.

Limitations

Several limitations should be considered when interpreting the findings of this study. First, although the holdout test set consisted entirely of real and previously unseen students, its small size ($n = 15$) limits statistical confidence and external validity; with each correctly classified student representing 6.67 percentage points of test accuracy, the confidence interval around the reported 93.33% accuracy is necessarily wide. Consequently, the reported performance metrics should be regarded as indicative rather than definitive and should not be generalised to institutions with substantially different student populations, assessment cultures, or GPA distributions without further validation.

A related concern is the extreme sparsity of the Hard class within the test set, which contained only a single Hard-class student. As a result, the apparently strong per-class metrics reported for this category are based on one observation and cannot provide a reliable estimate of population-level Hard-class performance. Additionally, the target labels were generated using a deterministic scoring rule derived from four of the five model input features. While grounded in formative assessment theory, these labels are not equivalent to genuine post-attempt student difficulty ratings and therefore introduce a degree of construct-feature overlap that creates a partially tautological relationship between predictors and outcomes. Although the framework supports authentic student ratings through the REAL_LABELS_PATH parameter, such data were not available for the present study.

The oversampling strategy used to address class imbalance introduces a further limitation, as five real Hard-class training students were upsampled to 37 observations, substantially increasing duplication, reducing within-class diversity, and potentially exaggerating the apparent separability of the Hard-class decision boundary. Consequently, models achieving Hard-class F1 scores of 1.00 should not be interpreted as having learned a robust or generalisable Hard-class representation. Finally, the LOWESS smoothing procedure was applied using row order as the x-axis rather than a temporally or theoretically meaningful variable, making it a pragmatic approximation rather than a fully grounded modelling choice. Access to time-stamped engagement and learning analytics data would enable a more theoretically justified smoothing approach in future implementations.

Future works

Future work should focus on several key extensions to strengthen and broaden the proposed framework. The most important enhancement is the collection of genuine post-attempt student difficulty ratings, replacing scoring-rule-derived labels to eliminate construct-feature overlap, enable true out-of-sample calibration validation, and better align model optimisation with the ultimate educational objective of accurately reflecting perceived difficulty. Expanding the study to larger, multi-institutional datasets involving hundreds or thousands of students would support the development of more expressive MLP architectures, facilitate robust fairness auditing across demographic subgroups, and provide statistically reliable confidence intervals for performance metrics.

Additionally, incorporating longitudinal engagement features derived from time-stamped LMS interaction logs, such as session frequency, time-on-task, forum participation, and assignment submission behaviour, could enrich learner profiles beyond cross-sectional survey indicators and capture evolving engagement patterns throughout an academic term. Another promising direction is the integration of the framework with Item Response Theory (IRT), combining profile-based difficulty priors with IRT-calibrated item parameters in a hybrid adaptive testing system that balances data-driven flexibility with psychometric rigour, as recommended in recent reviews of AI-enhanced adaptive assessment (Halkiopoulos and Gkintoni, 2024; Wang and Asif, 2025).

Finally, embedding the trained classifier within a learning management system to provide real-time adaptive question recommendations and continuously log student responses for model refinement would transform the current offline framework into a live adaptive assessment engine capable of ongoing self-improvement through online learning or periodic retraining.

Conclusion

This study demonstrates, at the level of an exploratory feasibility study, that undergraduate learner profiles, composed of readily obtainable academic and behavioural indicators, contain sufficient information to support machine-learning-based anticipatory difficulty classification at a level meaningfully above chance, and indicative of practical usefulness for adaptive question recommendation. However, this remains to be confirmed on larger, multi-institutional samples. The proposed MLP (RMSprop), operating with only 195 trainable parameters on an 8-dimensional feature vector, matched the performance of a well-regularised Logistic Regression at 93.33% holdout test accuracy (Wilson 95% CI: 70.2%–98.8%, $n = 15$), with misclassification patterns confined to adjacent difficulty levels

and a train-test gap of −3.56% consistent with genuine, if modest and not yet statistically conclusive, generalisation.

The framework's principal value lies not in the specific accuracy figures, which are subject to the constraints of a small single-institution dataset and the construct-feature overlap between the target variable and model inputs noted above, but in the methodological discipline applied throughout: splitting before oversampling, suppressing outliers without data loss, preserving ordinal feature structure, and regularising the neural network against a partially duplicated training set. These choices collectively ensure that the reported results represent real generalisation to real students rather than an artefact of data handling, although the small sample size means this should be read as a demonstration of feasibility and methodological soundness rather than as a definitive estimate of real-world performance; the small sample is not, by itself, a reason to dismiss the study, provided the conclusions drawn from it remain proportionate to the evidence available.

When extended to larger datasets with genuine post-attempt difficulty ratings, longitudinal engagement features, and integration with existing psychometric calibration frameworks, the approach described here offers a principled, interpretable, and educationally grounded foundation for next-generation personalised adaptive assessment systems, systems capable of meeting each learner at their current level and growing with them across an academic term.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Ethics statement

The studies involving humans were approved by La Trobe University, Human Ethics. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study; written informed consent obtained for each survey response.

Author contributions

AhA: Visualization, Writing – original draft, Data curation, Formal analysis, Methodology, Investigation, Conceptualization, Validation. BS: Validation, Writing – review & editing, Supervision. AL: Data curation, Writing – review & editing, Validation. AfA: Supervision, Writing – review & editing, Project administration. HS: Resources, Investigation, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/feduc.2026.1849701/full#supplementary-material

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